

Laws of Indices & Surds

Powers follow four laws; surds follow two. Almost every question here is a “no calculator” question — the working is the answer.



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THE TOPIC ON ONE MAP

Laws of Indices & Surds

01 — Indices

$$81^{\frac{1}{4}}$$

“in the form ax^n ”

$$a^{\frac{1}{3}} = -4$$

02 — Surds: simplify and expand

$$a\sqrt{b}$$

“show that”

03 — Rationalise the denominator

$$\frac{2}{\sqrt{5}-1}$$

$$a + b\sqrt{5}$$

04 — Equations with surd coefficients

$$x\sqrt{3} - 3 = x + \sqrt{3}$$

exact area

05 — Powers of x in the power

$$y = 3^x$$

$$p = 2^x$$

hence solve

01 Indices: Laws, Negative and Fractional Powers

Same base: add, subtract, multiply the powers. Negative means flip. The bottom of a fraction is the root.

WHEN YOU SEE → YOU DO

“write down the value of $81^{\frac{1}{4}}$ ”

→ root first, then power

“in the form ax^n ”, “ kx^n where k, n are simplified”

→ one number $\times$ one power of x

“given $y = \frac{1}{16}x^4$, express $y^{-\frac{3}{2}}$...”

→ number and x each get the power

“given $a^{\frac{1}{3}} = -4$, find a ”

→ undo: cube both sides

“use your answer to part (a)”

→ build the harder power from the easy one

$$a^m \times a^n = a^{m+n} \quad a^m \div a^n = a^{m-n} \quad (a^m)^n = a^{mn}$$

$$a^0 = 1 \quad a^{-n} = \frac{1}{a^n} \quad a^{\frac{m}{n}} = (\sqrt[n]{a})^m$$

WORKED MODEL • JUNE 2025

$p = \frac{1}{16}x^4$, $q = \frac{40}{x^3}$. Find pq^2 in the form kx^n .

1 $q^2 = 1600x^{-6}$

square every part

2 $\frac{1}{16} \times 1600 = 100$, $x^{4-6} = x^{-2}$

numbers, then powers

ANSWER $100x^{-2}$

IDEA 01

Practise: Indices

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Root first, then power

P1 • TOPIC 1 3 marks

Find the exact value of (a) $27^{\frac{2}{3}}$ (b) $16^{-\frac{3}{4}}$ (c) $\left(\frac{9}{4}\right)^{-\frac{1}{2}}$

Q2 **DR ESLAM** Form ax^n

P1 • TOPIC 1 3 marks

Write $\frac{6x^3 - 2\sqrt{x}}{4x^2}$ in the form $ax^p + bx^q$.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 1 3 marks

1 Write down the value of

(i) $81^{\frac{1}{4}}$

(ii) $8^{\frac{2}{3}}$

(iii) 5^{-2}

(3 marks)

Q4**PRACTICE**

Practice • Medium

P1 • TOPIC 1 4 marks

- 4 (a) Given that $y = \frac{1}{16}x^4$, express each of the following in the form ax^n , where a and n are constants.

$y^{\frac{1}{2}}$

(1 mark)

(b) y^{-1}

(1 mark)

(c) $y^{-\frac{3}{2}}$

(2 marks)

Q5**PRACTICE**

Practice • Very Hard

P1 • TOPIC 1 4 marks

- 4 (a) Given that $y = \frac{81}{16}x^{-12}$, express each of the following in the form ax^n , where a and n are constants.

$y^{\frac{3}{4}}$

(1 mark)

(b) $y^{-\frac{1}{2}}$

(1 mark)

(c) $\left(\frac{1}{y^2}\right)^{-3}$

(2 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 • October 2023 • Q2

P1 • TOPIC 1 4 marks

2. Given that

$$a = \frac{1}{64}x^2 \quad b = \frac{16}{\sqrt{x}}$$

express each of the following in the form kx^n where k and n are simplified constants.

(a) $a^{\frac{1}{2}}$ (1)

(b) $\frac{16}{b^3}$ (1)

(c) $\left(\frac{ab}{2}\right)^{-\frac{4}{3}}$ (2)

(Total for Question 2 is 4 marks)

Q7 **PAST PAPER** Pearson IAL WMA11 · June 2025 · Q1**P1 · TOPIC 1** **5 marks**

1. Given that

$$p = \frac{1}{16}x^4 \qquad q = \frac{40}{x^3}$$

express each of the following in the form kx^n where k and n are fully simplified constants.

(a) $p^{\frac{1}{2}}$ (1)

(b) $(pq)^{-1}$ (2)

(c) pq^2 (2)

(Total for Question 1 is 5 marks)

02 Surds: Simplify and Expand

Find the biggest square factor. Surds multiply like letters, and $\sqrt{a} \times \sqrt{a} = a$.

WHEN YOU SEE → YOU DO

“in the form $a\sqrt{b}$ ”

$$\rightarrow \sqrt{48} = \sqrt{16}\sqrt{3} = 4\sqrt{3}$$

“show that $2\sqrt{3} + 4\sqrt{12} - 3\sqrt{75} = -5\sqrt{3}$ ”

→ simplify every surd to the same $\sqrt{}$, then collect

“by expanding and simplifying, show that”

→ expand like brackets; $\sqrt{a}\sqrt{a} = a$

“ $\sqrt{a} + \sqrt{b} = \sqrt{a+b}$ is not true — give an example”

→ one numerical counter-example

WORKED MODEL • EXPAND

$$(\sqrt{12} - 3)(2 - \sqrt{75})$$

$$\textcircled{1} (2\sqrt{3} - 3)(2 - 5\sqrt{3})$$

simplify first

$$\textcircled{2} 4\sqrt{3} - 30 - 6 + 15\sqrt{3}$$

expand: $\sqrt{3}\sqrt{3} = 3$

$$\text{ANSWER } 19\sqrt{3} - 36$$

× **TRAP** $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$. $\sqrt{9+16} = 5$, but $\sqrt{9} + \sqrt{16} = 7$.

IDEA 02

Practise: Surds: Simplify and Expand

My questions first, then selected practice.

Q8 DR ESLAM Biggest square factor

P1 • TOPIC 1 3 marks

Simplify (a) $\sqrt{50} - \sqrt{18}$ (b) $(3 + \sqrt{2})(3 - \sqrt{2})$ **SELECTED PRACTICE****Q9** PRACTICE Practice • Easy

P1 • TOPIC 1 3 marks

3 Write the following in the form $a\sqrt{b}$

(i) $\sqrt{2} + \sqrt{8}$

(ii) $4\sqrt{3} - \sqrt{12} + 4\sqrt{48}$

(iii) $(2\sqrt{2})^3 + 3\sqrt{2}$

(3 marks)

Q10**PRACTICE**

Practice · Hard

P1 · TOPIC 1**5 marks**

5 (a) Show that $2\sqrt{3} + 4\sqrt{12} - 3\sqrt{75} = -5\sqrt{3}$

(3 marks)

(b) By expanding and simplifying, show that

$$(\sqrt{3} - 5)(5 - \sqrt{3}) = 10\sqrt{3} - 28$$

(2 marks)

Q11**PRACTICE**

Practice · Very Hard

P1 · TOPIC 1 12 marks

8 (a) Expand $(a + b\sqrt{5})^2$.

(2 marks)

(b) A square has an area of $(49 + 12\sqrt{5}) \text{ m}^2$ and a side length of $(a + b\sqrt{5}) \text{ m}$.

Show that $ab = 6$, and explain why this proves that a and b must both be non-negative.

(2 marks)

(c) Show that $a^4 - 49a^2 + 180 = 0$.

(3 marks)

(d) By using the substitution $y = a^2$ or otherwise, solve the equation $a^4 - 49a^2 + 180 = 0$.
Hence determine the side length of the square.

(5 marks)

03 Rationalise the Denominator

No surd may stay on the bottom. Multiply top and bottom by the **conjugate** — the same bracket with the sign flipped.

WHEN YOU SEE → YOU DO

“rationalise the denominator”, $\frac{3}{\sqrt{2}}$

$$\rightarrow \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$\frac{2}{3 + \sqrt{5}}$$

$$\rightarrow \times \frac{3 - \sqrt{5}}{3 - \sqrt{5}} \text{ — bottom becomes } 9 - 5$$

“in the form $a + b\sqrt{5}$ where a, b are rational”

→ rationalise, then split the fraction

1

Conjugate

Flip the sign in the middle of the bottom.

2

Bottom

$$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b \text{ — no surd left.}$$

3

Top

Expand, then split into $p + q\sqrt{b}$.

WORKED MODEL • JUNE 2022

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}}$$

$$1 \times \frac{7 + 3\sqrt{5}}{7 + 3\sqrt{5}}$$

conjugate

$$2 \text{ bottom: } 49 - 45 = 4; \text{ top: } 28\sqrt{5} + 60 - 35 - 15\sqrt{5}$$

expand

ANSWER $\frac{25}{4} + \frac{13}{4}\sqrt{5}$

IDEA 03

Practise: Rationalise the Denominator

My questions first, then selected practice, then real past papers.

Q12 DR ESAM Conjugate

P1 · TOPIC 1 3 marks

Rationalise (a) $\frac{6}{\sqrt{3}}$ (b) $\frac{4}{\sqrt{5}-1}$

SELECTED PRACTICE**Q13** PRACTICE Practice · Easy

P1 · TOPIC 1 2 marks

4 Rationalise the denominator of $\frac{3}{\sqrt{2}}$.

(2 marks)

Q14 **PRACTICE** Practice · Easy

P1 • TOPIC 1 3 marks

9 Simplify

$$\frac{3 - 2\sqrt{3}}{4 - \sqrt{3}}$$

(3 marks)

Q15 **PRACTICE** Practice · Very Hard

P1 • TOPIC 1 5 marks

6 (a) $\sqrt{a} - \sqrt{b} = \sqrt{a-b}$ is not true in general. Give an example of an a and a b for which it is true.

(1 mark)

(b) Show that $\frac{2 - \sqrt{3}}{1 + \sqrt{3}} = a + b\sqrt{3}$, where a and b are rational numbers.

(4 marks)

PAST PAPERS

Q16 PAST PAPER Pearson IAL WMA11 • June 2022 • Q3

P1 • TOPIC 1 5 marks

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

- (i) Show that $\frac{\sqrt{180} - \sqrt{80}}{\sqrt{5}}$ is an integer and find its value.

(2)

- (ii) Simplify

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}}$$

giving your answer in the form $a + b\sqrt{5}$ where a and b are rational numbers.

(3)

(Total 5 marks)

Q17 PAST PAPER Pearson IAL WMA11 · June 2023 · Q6

P1 · TOPIC 1 7 marks

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Expand and simplify

$$\left(r - \frac{1}{r}\right)^2$$

(2)

(b) Express $\frac{1}{3 + 2\sqrt{2}}$ in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

(c) Use the results of parts (a) and (b), or otherwise, to show that

$$\sqrt{3 + 2\sqrt{2}} - \frac{1}{\sqrt{3 + 2\sqrt{2}}} = 2$$

(3)

(Total for Question 6 is 7 marks)

Q18**PAST PAPER**

Pearson IAL WMA11 • October 2023 • Q3

P1 • TOPIC 1 6 marks

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Write $\frac{8 - \sqrt{15}}{2\sqrt{3} + \sqrt{5}}$ in the form $a\sqrt{3} + b\sqrt{5}$ where a and b are integers to be found.

(3)

(b) Hence, or otherwise, solve

$$(x + 5\sqrt{3})\sqrt{5} = 40 - 2x\sqrt{3}$$

giving your answer in simplest form.

(3)

(Total for Question 3 is 6 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 · October 2024 · Q2

P1 · TOPIC 1 7 marks

2. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i) Simplify fully

$$\frac{3y^3(2x^4)^3}{4x^2y^4} \quad (3)$$

(ii) Find the exact value of a such that

$$\frac{16}{\sqrt{3} + 1} = a\sqrt{27} + 4$$

Write your answer in the form $p\sqrt{3} + q$ where p and q are fully simplified rational constants.

(4)

(Total for Question 2 is 7 marks)

04 Equations with Surd Coefficients

Treat $\sqrt{3}$ like a number: collect the x terms, factorise x out, divide — then rationalise.

WHEN YOU SEE → YOU DO

“solve $x\sqrt{3} - 3 = x + \sqrt{3}$, answer in the form $p + q\sqrt{3}$ ”

→ collect, factorise, divide, rationalise

area / length with surds (“exact value”)

→ set up the formula, divide, rationalise

“(a + b $\sqrt{5}$)² = 49 + 12 $\sqrt{5}$ ”

→ compare rational parts and surd parts

WORKED MODEL • JUNE 2024

$$x\sqrt{3} - 3 = x + \sqrt{3}$$

$$1 \quad x\sqrt{3} - x = 3 + \sqrt{3}$$

collect x

$$2 \quad x(\sqrt{3} - 1) = 3 + \sqrt{3} \Rightarrow x = \frac{3 + \sqrt{3}}{\sqrt{3} - 1}$$

factorise, divide

$$3 \quad x \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} : \frac{6 + 4\sqrt{3}}{2}$$

rationalise

$$\text{ANSWER} \quad x = 3 + 2\sqrt{3}$$

IDEA 04

Practise: Equations with Surds

My questions first, then selected practice, then real past papers.

Q20 **DR ESLAM** Factorise x out**P1 · TOPIC 1** 4 marks

Solve $x\sqrt{2} + 3 = 2x + \sqrt{2}$, giving x in the form $a + b\sqrt{2}$.

$x =$ _____

SELECTED PRACTICE**Q21** **PRACTICE** Practice · Medium**P1 · TOPIC 1** 4 marks

7 Solve the equation $6 - x\sqrt{7} = \frac{2x}{\sqrt{7}}$, giving your answer in the form $\frac{a\sqrt{7}}{b}$ where a and b are integers.

(4 marks)

Q22**PRACTICE**

Practice · Hard

P1 · TOPIC 1**4 marks**

- 7 Solve the equation $66 + x\sqrt{75} = \frac{4x}{\sqrt{3}}$, giving your answer in the form $a\sqrt{b}$ where a and b are integers.

(4 marks)**Q23****PRACTICE**

Practice · Very Hard

P1 · TOPIC 1**5 marks**

- 7 Solve the equation $\sqrt{20} + \frac{\sqrt{5}}{2x} = \frac{1}{x\sqrt{45}}$

(5 marks)

PAST PAPERS

Q24 PAST PAPER Pearson IAL WMA11 · January 2022 · Q3

P1 · TOPIC 1 7 marks

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i)

$$f(x) = (x + \sqrt{2})^2 + (3x - 5\sqrt{8})^2$$

Express $f(x)$ in the form $ax^2 + bx\sqrt{2} + c$ where a , b and c are integers to be found. (3)

(ii) Solve the equation

$$\sqrt{3}(4y - 3\sqrt{3}) = 5y + \sqrt{3}$$

giving your answer in the form $p + q\sqrt{3}$ where p and q are simplified fractions to be found. (4)

(Total 7 marks)

Q25 **PAST PAPER** Pearson IAL WMA11 • June 2024 • Q2**P1 • TOPIC 1** **6 marks**

2. (i) Given that $m = 2^n$, express each of the following in simplest form in terms of m .

(a) 2^{n+3}

(1)

(b) 16^{3n}

(2)

- (ii) **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

Solve the equation

$$x\sqrt{3} - 3 = x + \sqrt{3}$$

giving your answer in the form $p + q\sqrt{3}$ where p and q are integers.

(3)

(Total for Question 2 is 6 marks)

05 Powers of x in the Power: Substitute

9^x , 3^{x+2} , 3^{x-1} all hide one letter, $p = 3^x$. Rewrite everything in p and it becomes a quadratic.

WHEN YOU SEE → YOU DO

“given $y = 3^x$, express 3^{3x} ... in terms of y ”

$$\longrightarrow 3^{3x} = (3^x)^3 = y^3$$

“by substituting $p = 2^x$, show that”

$$\longrightarrow 4^x = p^2, \quad 2^{x+3} = 8p, \quad 2^{x-1} = \frac{p}{2}$$

“hence solve”

$$\longrightarrow \text{solve for } p, \text{ then } 2^x = p \rightarrow x$$

p negative

$$\longrightarrow \text{reject: } 3^x > 0 \text{ always}$$

WORKED MODEL • JANUARY 2024

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4, \quad p = 2^x$$

$$1 \quad 2p^2 - 8p = \frac{17}{2}p - 4$$

rewrite in p

$$2 \quad 4p^2 - 33p + 8 = 0 \Rightarrow p = 8 \text{ or } \frac{1}{4}$$

quadratic

$$\text{ANSWER} \quad 2^x = 8 \Rightarrow x = 3, \quad 2^x = \frac{1}{4} \Rightarrow x = -2$$

IDEA 05

Practise: Substitute $p = 3^x$

My questions first, then real past papers.

Q26 DR ESLAM Substitute

P1 · TOPIC 1 4 marks

By substituting $p = 3^x$, solve $9^x - 10 \times 3^x + 9 = 0$. $x =$ _____

PAST PAPERS

Q27 PAST PAPER Pearson IAL WMA11 • January 2020 • Q2

P1 • TOPIC 1 5 marks

2. Given $y = 3^x$, express each of the following in terms of y . Write each expression in its simplest form.

(a) 3^{3x} (1)

(b) $\frac{1}{3^{x-2}}$ (2)

(c) $\frac{81}{9^{2-3x}}$ (2)

(Total 5 marks)

Q28 **PAST PAPER** Pearson IAL WMA11 · January 2023 · Q5**P1 · TOPIC 1** **6 marks**

5. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) By substituting $p = 3^x$, show that the equation

$$3 \times 9^x + 3^{x+2} = 1 + 3^{x-1}$$

can be rewritten in the form

$$9p^2 + 26p - 3 = 0 \quad (3)$$

(b) Hence solve

$$3 \times 9^x + 3^{x+2} = 1 + 3^{x-1} \quad (3)$$

(Total for Question 5 is 6 marks)

Q29**PAST PAPER**

Pearson IAL WMA11 · January 2024 · Q4

P1 · TOPIC 1 6 marks

4. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) By substituting $p = 2^x$, show that the equation

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4$$

can be written in the form

$$4p^2 - 33p + 8 = 0 \quad (3)$$

(b) Hence solve

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4 \quad (3)$$

(Total for Question 4 is 6 marks)

Q30**PAST PAPER**

Pearson IAL WMA11 · January 2025 · Q4

P1 · TOPIC 1 9 marks

4. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) Given that

$$y = a^x \quad \text{where } a \text{ is a positive constant}$$

express, in simplest form, in terms of y and a

(a) a^{3x+1} (1)

(b) $\frac{5}{(3a^{1-x})^{-2}}$ (3)

- (ii) (a) Use the substitution $p = 9^t$ to show that the equation

$$3(3^{4t+2} + 1) = 82 \times 9^t$$

can be rewritten as

$$27p^2 - 82p + 3 = 0$$
 (2)

- (b) Hence solve

$$3(3^{4t+2} + 1) = 82 \times 9^t$$
 (3)

(Total for Question 4 is 9 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|---|
| 01 | Indices | Add · subtract · multiply powers. Negative = flip, fraction = root. |
| 02 | Surds | Biggest square factor. $\sqrt{a}\sqrt{a} = a$. $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$. |
| 03 | Rationalise | × conjugate. Bottom: $a^2 - b$. |
| 04 | Surd equations | Collect, factorise x , divide, rationalise. |
| 05 | $p = 3^x$ | Rewrite in $p \rightarrow$ quadratic $\rightarrow$ back to x . Reject $p \leq 0$. |

BOTTOM LINE Same base $\rightarrow$ laws · biggest square factor $\rightarrow$ simplify · surd on the bottom $\rightarrow$ conjugate · power in the power $\rightarrow$ substitute.

Answers

Laws of Indices & Surds • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • INDICES

- Q1** Dr Eslam • Root first, then power (a) 9 (b) $\frac{1}{8}$ (c) $\frac{2}{3}$
- Q2** Dr Eslam • Form ax^n $\frac{3}{2}x - \frac{1}{2}x^{-\frac{3}{2}}$
- Q3** Practice • Easy (i) 3 (ii) 4 (iii) $\frac{1}{25}$
- Q4** Practice • Medium (a) $\frac{1}{4}x^2$ (b) $16x^{-4}$ (c) $64x^{-6}$
- Q5** Practice • Very Hard (a) $\frac{27}{8}x^{-9}$ (b) $\frac{4}{9}x^6$ (c) $\frac{64}{729}x^{18}$
- Q6** Pearson IAL WMA11 • October 2023 • Q2 (a) $\frac{1}{8}x$ (b) $\frac{1}{256}x^{\frac{3}{2}}$ (c) $16x^{-2}$
- Q7** Pearson IAL WMA11 • June 2025 • Q1 (a) $\frac{1}{4}x^2$ (b) $\frac{2}{5}x^{-1}$ (c) $100x^{-2}$

IDEA 02 • SURDS: SIMPLIFY AND EXPAND

- Q8** Dr Eslam • Biggest square factor (a) $2\sqrt{2}$ (b) 7
- Q9** Practice • Easy (i) $3\sqrt{2}$ (ii) $18\sqrt{3}$ (iii) $19\sqrt{2}$
- Q10** Practice • Hard (a) $2\sqrt{3} + 8\sqrt{3} - 15\sqrt{3}$ (b) $5\sqrt{3} - 3 - 25 + 5\sqrt{3}$
- Q11** Practice • Very Hard (a) $a^2 + 5b^2 + 2ab\sqrt{5}$ (d) $a^2 = 4$ or 45; side $2 + 3\sqrt{5}$ m

IDEA 03 • RATIONALISE THE DENOMINATOR

- Q12** Dr Eslam • Conjugate (a) $2\sqrt{3}$ (b) $1 + \sqrt{5}$
- Q13** Practice • Easy $\frac{3\sqrt{2}}{2}$
- Q14** Practice • Easy $\frac{6-5\sqrt{3}}{13}$
- Q15** Practice • Very Hard (a) e.g. $b = 0$ (b) $-\frac{5}{2} + \frac{3}{2}\sqrt{3}$
- Q16** Pearson IAL WMA11 • June 2022 • Q3 (i) $6 - 4 = 2$ (ii) $\frac{25}{4} + \frac{13}{4}\sqrt{5}$
- Q17** Pearson IAL WMA11 • June 2023 • Q6 (a) $r^2 - 2 + r^{-2}$ (b) $3 - 2\sqrt{2}$ (c) $\sqrt{3 + 2\sqrt{2}} = 1 + \sqrt{2}$
- Q18** Pearson IAL WMA11 • October 2023 • Q3 (a) $3\sqrt{3} - 2\sqrt{5}$ (b) $x = 15\sqrt{3} - 10\sqrt{5}$
- Q19** Pearson IAL WMA11 • October 2024 • Q2 (i) $\frac{6x^{10}}{y}$ (ii) $a = \frac{8}{3} - \frac{4}{3}\sqrt{3}$

IDEA 04 • EQUATIONS WITH SURDS

- Q20** Dr Eslam • Factorise x out $x(\sqrt{2} - 2) = \sqrt{2} - 3 \Rightarrow x = 2 + \frac{\sqrt{2}}{2}$
- Q21** Practice • Medium $x = \frac{2\sqrt{7}}{3}$
- Q22** Practice • Hard $x = -6\sqrt{3}$
- Q23** Practice • Very Hard $x = -\frac{13}{60}$
- Q24** Pearson IAL WMA11 • January 2022 • Q3 (i) $10x^2 - 58x\sqrt{2} + 202$ (ii) $y = \frac{57}{23} + \frac{41}{23}\sqrt{3}$
- Q25** Pearson IAL WMA11 • June 2024 • Q2 (i) $8m, m^{12}$ (ii) $x = 3 + 2\sqrt{3}$

IDEA 05 • SUBSTITUTE $p = 3^x$

- Q26** Dr Eslam • Substitute $p = 1$ or 9 : $x = 0$ or $x = 2$
- Q27** Pearson IAL WMA11 • January 2020 • Q2 (a) y^3 (b) $\frac{9}{y}$ (c) y^6
- Q28** Pearson IAL WMA11 • January 2023 • Q5 (b) $p = \frac{1}{9}$ ($p = -3$ rejected): $x = -2$
- Q29** Pearson IAL WMA11 • January 2024 • Q4 (b) $x = 3$ or $x = -2$
- Q30** Pearson IAL WMA11 • January 2025 • Q4 (i) $ay^3; \frac{45a^2}{y^2}$ (ii) $p = 3$ or $\frac{1}{27}$: $t = \frac{1}{2}$ or $-\frac{3}{2}$